

PSYC 51.17: Models of Language and Communication - Week 4

Cognitive Models¹ of Semantic Representation

Lecture 14: How Humans Represent Meaning

PSYC 51.17: Models of Language and Communication - Week 4

Winter 2026

Winter 2026

Today's Lecture

1. **Human vs. Computational Semantics**
2. **Cognitive Theories of Meaning**
3. **Embodied & Grounded Cognition**
4. **Semantic Similarity: What Does It Mean?**
5. **Empirical Evidence from Cognitive Science**
6. **Bridging the Gap: Models Meet Minds**

*Goal: Understand what computational models are
really learning*

Winter 2026

The Fundamental Question

What does "meaning" actually mean?

Computational View:

- Vectors in high-dimensional space
- Learned from text co-occurrence
- Distributional patterns

Human View:

- Sensory experiences
- Emotional associations
- Physical grounding
- Social context
- Multimodal integration

How Do Humans Represent Meaning?

Example: The word "coffee"

What comes to YOUR mind?

- **Visual:** brown liquid, mug, steam, beans
- **Olfactory:** aroma, roasted smell
- **Gustatory:** bitter taste, smooth texture
- **Tactile:** hot, warm cup, liquid
- **Auditory:** brewing sounds, pouring
- **Motor:** lifting cup, drinking motion
- **Contextual:** morning routine, work, café
- **Emotional:** comfort, alertness, pleasure
- **Social:** conversations, meetings

Word2Vec representation:

```
coffee = [0.23, -0.45, 0.67,  
           0.12, -0.89, ... ]
```

(300 numbers learned from text)

What's missing?

- No sensory grounding
- No embodied experience
- No emotional content
- No perceptual features
- No action affordances

The Symbol Grounding Problem

Harnad (1990): Can symbols have intrinsic meaning?

The Chinese Room (Searle, 1980):

```
1 | Input:
2 |   ↓
3 | [Rule Book: If see
4 |   output ]
5 |   ↓
6 | Output:
7 |
8 | Correct response! But no
9 | understanding of Chinese.
```

Analogy to LLMs:

```
1 | Input: "Is a penguin a bird?"
```

Symbol Systems:

- Symbols refer to other symbols
- Purely syntactic manipulation
- No connection to world

Grounded Systems:

- Symbols connected to perception
- A child learns "dog" by:
- SEEING dogs
- PETTING dogs

Embodied Cognition Theory

Meaning arises from bodily experience and sensorimotor interaction

Key Principles:

1. **Embodiment:** Cognition shaped by body
2. **Situatedness:** Meaning context-dependent
3. **Enactivism:** Knowing through doing
4. **Grounding:** Concepts tied to perception/action

Conceptual Metaphors (Lakoff & Johnson):

- | | |
|---|---|
| 1 | Physical → Abstract mapping: |
| 2 | |
| 3 | "WARM personality" ← holding warm drink |
| 4 | primes positive judgments! |
| 5 | |
| 6 | "HIGH status" ← up = good, |

The Distributional Hypothesis Revisited

"You shall know a word by the company it keeps"

--- J.R. Firth (1957)

Strong version: Word meaning IS distributional patterns

Weak version: Distributional patterns REFLECT meaning

Supports:

- Works remarkably well in practice
- Captures semantic similarity
- Enables analogical reasoning
- Scales to huge vocabularies

Limitations:

- Correlation $\neq$ causation
- Lacks perceptual grounding
- No embodied understanding
- Missing common sense

What Does "Semantic Similarity" Really Mean?

Different types of similarity:

Taxonomic (IS-A):

- 1 | dog cat: Both are animals
- 2 | Similarity: HIGH

Thematic (GOES-WITH):

- 1 | dog leash: Co-occur in events
- 2 | Relatedness: HIGH
- 3 | Similarity: LOW!

Test Yourself:

- 1 | Which is more SIMILAR to

What Word2Vec Says:

- 1 | # Word2Vec often gets this wrong!
- 2 | model.similarity('coffee', 'cup') # 0.65
- 3 | model.similarity('coffee', 'tea') # 0.62
- 4 |
- 5 | # Cup ranked higher due to co-occurrence!
- 6 | # But tea is categorically more similar

8

SimLex-999 vs WordSim-353:

Semantic Projection: Recovering Human Knowledge

Grand et al. (2022): Can we extract human-like features from embeddings?

The Experiment:

1. Collect human ratings on perceptual features

1	\$[0.23, -0.45, ..., 0.12] → Perceptual Features →	9
---	--	---

Multimodal Models: Bridging the Gap

Combining language with perception

Vision-Language Models:

- CLIP (OpenAI)
- ALIGN (Google)
- Flamingo (DeepMind)
- GPT-4V (OpenAI)

1

Vision →
Encoder →
Text →
Encoder →
Similar!

10

Benefits:

Conceptual Spaces Theory

Gärdenfors (2000): Meaning as geometry

Key Ideas:

- Concepts represented in quality dimensions
- Dimensions: color, size, temperature, etc.
- Each dimension has a metric

Relation to Embeddings:

- Similar geometric structure
- But learned dimensions
- Not interpretable qualities₁₁
- No grounding in perception

Lexical Semantic Theories

How do linguists think about word meaning?

1. Feature-Based:

- Words = bundles of features
- bachelor = [+human, +male, +adult, -married]
- Compositional
- Logical

2. Prototype Theory:

- Categories have best examples
- Robin is a prototypical bird

4. Construction Grammar:

- Meaning from form-function pairings
- Idioms, patterns
- Usage-based

5. Word Sense Disambiguation:

- Words have multiple senses
- WordNet: hierarchical ontology
- Bank₁ (financial), Bank₂ (riverside)

Empirical Evidence from Neuroscience

What does the brain tell us about semantic representation?

fMRI Studies:

- Can predict brain activity from word embeddings
- Semantic information distributed across cortex
- Different regions for different features
- Temporal lobe: objects
- Motor cortex: actions
- Visual cortex: visual features

Findings:

Brain vs. Model Representations:

Distributed		
Hierarchical		
Context-sensitive		(BERT)
Grounded		
Fast		

Key Insight:

- Models capture some aspects
- But miss crucial grounding
- Convergent representation learning?

The "Stochastic Parrots" Debate

Bender et al. (2021): On the Dangers of Stochastic Parrots

The Argument:

- LLMs learn form, not meaning
- "Stochastic parrots" - repeating patterns
- No understanding of world
- No communicative intent
- Risk: Mistaking fluency for understanding

Evidence:

- Fail on simple reasoning
- Sensitive to phrasing

Counter-Arguments:

- Emergent capabilities at scale
- Performance on complex tasks
- Transfer to new domains
- Maybe understanding = prediction?
- Pragmatic success criterion

The Core Question

Can meaning arise from form alone?

Or do we need grounding in:

Common Sense Reasoning

What humans know but models don't

Physical Intuition Failures:

```
1 | Q: "Can you fit an elephant
2 |   in a refrigerator?"
3 |
4 | GPT-3: "Yes, if you open the
5 |   door wide enough ... "
```

Winograd Schema (reasoning):

```
1 | "The trophy doesn't fit in the
2 |   brown suitcase because it is
3 |   too [small/large]."
```

4 |

```
5 | What does "it" refer to?
```

Social Intuition Failures:

```
1 | Q: "John told Mary he loved her.
2 |   How did Mary feel?"
3 |
4 | Depends on context:
5 | - First date? → Surprised/happy
6 | - After argument? → Relieved
7 | - Unwanted? → Uncomfortable
8 |
9 | Models miss social nuance!
```

15

Why Models Struggle:

- Physical/social knowledge rarely stated

Compositionality: Phrases and Sentences

How do we combine word meanings?

The Problem:

```
1 # Vector math doesn't work!  
2 vec("hot") + vec("dog") ≠  
  vec("hot dog")  
3  
4 # "hot dog" = food item  
5 # "hot" + "dog" = warm canine  
6  
7 # Same issue:  
8 vec("red") + vec("herring") ≠  
  vec("red herring")  
9 # red herring = distraction,  
  not a fish!
```

What Transformers Learn:

```
1 Input: "break a leg"  
2 ↓  
3 Attention sees this phrase  
4 often in "good luck" contexts  
5 ↓  
6 Output: idiomatic meaning  
7  
8 But fails on novel  
  combinations!
```

16

The Negation Problem:

```
1 # BERT struggles with negation
```

Grand Discussion

Do large language models "understand" language?

Arguments FOR:

- Solve complex tasks
- Generalize to new domains
- Show emergent capabilities
- Capture linguistic structure
- Pragmatic criterion: if it works...
- Maybe understanding = prediction
- Human understanding also imperfect

*"The question is not whether machines think,
but whether they behave intelligently" - Turing*

Arguments AGAINST:

- No grounding in reality
- No embodied experience
- No intentionality
- Brittle, exploit shortcuts
- Hallucinate confidently
- No causal reasoning
- Missing common sense
- Form without meaning

"Understanding requires grounding in

- perceptual grounding
2. **Embodied Cognition:** Human meaning tied to bodily experience
 3. **Distributional Semantics:** Powerful but incomplete theory
 4. **Semantic Similarity:** Multiple types, models capture some
 5. **Empirical Evidence:** Models align with neural patterns but miss multimodality
 6. **Common Sense:** Models struggle with physical and social reasoning
 7. **Compositionality:** Non-literal language remains

- Barsalou, L. (2008). "Grounded Cognition"
- Gärdenfors, P. (2000). "Conceptual Spaces: The Geometry of Thought"

Distributional Semantics:

- Firth, J.R. (1957). "A Synopsis of Linguistic Theory"
- Boleda, G. (2020). "Distributional Semantics and Linguistic Theory"
- Hill et al. (2015). "SimLex-999"

Critical Perspectives:

Questions?

Next Week:

Advanced Topics in Language Models

Scaling, emergent abilities, and the future of NLP

Winter 2026